NeblinIA-Speech

A foundational ASR system for Mexican Indigenous languages

Technical Report, preview-0.1

NeblinIA Lab. 2026-06-20.

Abstract. We build a foundational automatic speech recognition (ASR) model for 23 Mexican Indigenous languages plus Mexican Spanish, starting from Whisper-large-v3-turbo and roughly ten hours of open data per language. We report a final fair-evaluation word error rate (WER) of 58.99 and character error rate (CER) of 26.45 on a private, contamination-resistant benchmark, a reduction of 7.0 WER points over our prior best. The path that worked combined broad multilingual data scaling with reinforcement learning (GSPO). Equally important, we document the methods that did not work: teacher-forced supervised fine-tuning that overfits and amplifies repetition, full fine-tuning that is incompatible with our toolchain, label smoothing that crashes the model, best-of-K self-distillation that overfits, maximum-entropy frontier weighting that fails on bimodal data, and anti-repetition decode guards that hurt because these languages use grammatical reduplication. The central methodological finding is that teacher-forced validation metrics anti-correlate with real autoregressive quality in late training, so checkpoint selection must use autoregressive evaluation.

Introduction

NeblinIA is a foundational research lab focused on speech technology for the Indigenous languages of Mexico. This report covers the first preview model, an end-to-end effort to take a general multilingual ASR backbone and adapt it to 23 low-resource Mexican languages spanning the Oto-Manguean family (Mixtec, Zapotec, Chinantec, Amuzgo, Mazatec, Cuicatec), Uto-Aztecan (Nahuatl), Totonacan (Totonac), and Mixe-Zoque (Zoque), among others. These languages are polysynthetic or agglutinative, frequently tonal, and lack standardized orthographies, which makes word error rate a harsh metric and makes the data scarcity acutely limiting.

The stated target was an average WER of 20 with no repetition. We did not reach 20, and a core contribution of this report is an honest account of why: at roughly ten hours per language, a handful of hard or data-starved languages dominate the average and cannot be fixed by modeling alone. We did reach a usable, top-ranked model, and we mapped the solution space thoroughly enough that the remaining levers are clear.

Goals and constraints

The lab operates under three standing constraints that shaped every decision. First, the benchmark must be contamination-resistant and evaluated fairly, meaning every model (ours and the baselines) is scored with the identical decoding protocol. Second, all training data must be openly licensed. Third, the work must be reproducible, version controlled, and documented, including the failures.

Data

In-domain corpus

The in-domain training and evaluation data come from the Omnilingual ASR corpus (`facebook/omnilingual-asr-corpus`, CC BY 4.0). For the 23 target Mexican languages this yields 22,549 training segments and 5,784 development segments after forced alignment of the roughly 60-second source clips into segments under 30 seconds. The per-language counts are balanced near 1,000 segments, with one exception (`nhq` at 258 segments) that proves important later. This is the hard ceiling of the in-domain source: about ten hours per language.

Open transfer data

To scale beyond the in-domain ceiling we added Common Voice version 26 (CC0) for ten related Mexican Indigenous languages already present on disk: Nahuatl variants `ncx` and `nlv`, Zoque `zoc` (same family as our test language `zoh`), Mazatec `mau`, Cuicatec `cut` and `cux`, Purépecha `pua`, Yaqui `yaq`, Seri `sei`, and Tarahumara `tar`. These contribute about 75,000 additional validated clips. Common Voice clips are short single utterances, so they need no forced alignment, and `soundfile` reads the mp3 audio directly. We kept the checkpoint commercial-clean by excluding all CC-BY-NC sources (for example the large OpenSLR Mixtec and Nahuatl deposits).

The combined "broad" manifest holds 97,646 clips. All audio is decoded under a single forced `es` language token, the standard low-resource bucket trick: the token becomes a generic "Mexican languages" marker and the model learns to map acoustics to the target orthography directly.

Source	License	Clips	Role
Omnilingual ASR (23 MX langs)	CC BY 4.0	22,549	in-domain train, plus 5,784 dev
Common Voice v26 (10 related)	CC0	~75,000	transfer fuel, train only
Broad combined manifest	open	97,646	multistage pre-training

Benchmark and evaluation

The fair protocol

The held-out test set (MEXA, 5,925 clips across the 23 languages plus Spanish) is private: its reference transcripts are never published, so it functions as a contamination-resistant answer key. Every model is scored through the same faster-whisper (CTranslate2) engine with beam size 1, faster-whisper temperature fallback, and automatic language detection for the non-Spanish languages. This parity is the integrity of the leaderboard. An earlier internal result that gave our model a friendlier decode than the baselines was discarded once the unfairness was identified.

The teacher-forced trap

The single most important evaluation lesson of the project: the in-training validation metric is teacher-forced (the decoder always sees the gold prefix), and it anti-correlates with real autoregressive quality in late training. A model can show a beautiful, monotonically improving teacher-forced WER while its free-running greedy decoding collapses into repetition. Because of this, we built a fast autoregressive triage that runs raw greedy decoding (single temperature, no fallback) on the development set and reports per-language WER, CER, and a loop rate (fraction of clips where a 3-gram repeats three or more times in a row). All checkpoint selection uses this autoregressive triage, not the teacher-forced number.

Methods

Supervised fine-tuning

The backbone is Whisper-large-v3-turbo (809M parameters, a distilled model with a four-layer decoder), adapted with LoRA through the Unsloth library. We extended the adapter from the minimal attention projections to all linear modules (`q`, `k`, `v`, `out`, `fc1`, `fc2`), at rank 64, with gradient checkpointing. The decisive hyperparameter turned out to be not a learning rate but a stopping rule: training must stop early, near 0.9 of an epoch, because full training overfits and amplifies repetition.

Reinforcement learning (GSPO)

The post-training method is Group Sequence Policy Optimization (GSPO, arXiv 2507.18071), a critic-free, group-relative policy gradient. For each clip we sample a group of eight hypotheses, score each with a verifiable reward, compute a group-relative advantage (reward minus group mean), and update with a sequence-level, length-normalized importance ratio and a small Kullback-Leibler penalty to a frozen supervised anchor. The reward is a composite of negative CER and negative WER plus explicit anti-repetition penalties for over-generation length and n-gram repetition. Whisper consumes audio, so the loop is hand rolled rather than borrowed from a text-only library; a key efficiency fix encodes the audio once per clip and reuses the encoder output across the eight samples to avoid an out-of-memory failure.

Results

The final model, internally `preview-0.9-broadgspo` and released as NeblinIA-Speech preview-0.1, scores as follows on the fair benchmark.

Model	WER	CER
NeblinIA-Speech preview-0.1 (broad data plus GSPO)	**58.99**	**26.45**
Prior best (preview-0.3, attention-only SFT plus GSPO)	66.02	28.26

The improvement is 7.0 WER points and the model evaluates quickly, with no catastrophic looping. Per-language results are bimodal, which is the heart of the remaining problem: a group of languages is approaching usable quality while a group of hard or data-starved languages still fails.

Lang	WER	Lang	WER	Lang	WER
spa	18.8	vmj	66.4	nhn	77.2
zor	39.2	vmc	66.5	chq	77.3
zoh	49.6	ztn	67.3	nhg	79.7
tlp	53.1	ztu	69.2	mig	86.2
amu	56.4	tcf	74.8	xti	88.8
trq	60.0	vmz	75.9	nhq	100.2
ztp	61.8	zpv	76.1	zts	106.4
vmp	62.3	pmq	76.6	ncf	65.6

Spanish at 18.8 and Zapotec `zor` at 39.2 show that the per-language floor is genuinely low when data exists. The languages above 85 (`mig`, `xti`, `nhq`, `zts`) are the ones that loop and that lack the data to be fixed by the current recipe. `nhq`, with only 258 training segments, is the clearest case of data starvation.

What worked, measured autoregressively

The autoregressive development triage (raw greedy, 460 clips) is the metric that tracks reality. It tells a clean story: data scale reduces looping, but reinforcement learning is what closes the gap.

Configuration	WER	CER	Loop rate
preview-0.3: attention-only SFT, then GSPO	89	50	11.3%

Configuration	WER	CER	Loop rate
broad base: 97k data, all-module SFT, 0.9 epoch	108	60	15.9%
p05: 22.5k data, all-module SFT, ~4 epochs	139	87	26.3%
balanced-broad: 47.5k data, all-module SFT, 2 epochs	123	75	23.9%

Reading across: tripling the data (p05 to broad) cut the loop rate from 26.3% to 15.9% and WER from 139 to 108, so data scale helps free-running generation. But the supervised base still loses to the earlier GSPO model, because reinforcement learning is decisive. Applying GSPO to the broad base drove held-out greedy dev CER from 0.544 to 0.422 and produced the final 58.99 fair-eval result.

What failed

This section is the point of the report. Each item below is a real experiment that did not work, with the reason.

Best-of-K self-distillation (RFT)

We sampled eight hypotheses per clip and kept the lowest-CER sample as a new supervised target (the rejection-sampling fine-tuning recipe). The model overfit its own low-diversity samples and amplified looping rather than reducing it. The recovered checkpoint looped so badly that a fair evaluation did not finish in 33 minutes (a clean evaluation takes about five). Negative result, abandoned.

All-module SFT alone

Extending the adapter to all linear modules improved the teacher-forced WER from 78 to 64, which looked excellent. The autoregressive triage told the truth: WER 139 and a 26.3% loop rate, worse than the baseline. More adapter capacity bought a better teacher-forced fit and a worse free-running model, the signature of exposure bias.

Full fine-tuning and label smoothing

Unsloth full fine-tuning of Whisper crashes at the first step with a decoder input conflict (`cannot specify both decoder_input_ids and decoder_inputs_embeds`). Separately, and more insidiously, label smoothing triggers the identical crash because the Hugging Face label smoother feeds the model differently from the patched Whisper forward. We misdiagnosed this twice (first blaming full fine-tuning, then a corrupted compile cache) before isolating label smoothing as the true cause. The fix is to use neither with this toolchain.

Maximum-entropy frontier weighting (MGPO)

The VibeThinker signal phase weights each clip by how close it sits to the learnable frontier (solved about half the time). This assumes a smooth difficulty gradient. Our difficulty distribution is bimodal: clips are either solved or total looping garbage, with almost nothing in between. The frontier was nearly empty, so the weighting down-weighted essentially everything and the gradient went dead. The VibeThinker recipe is built for reasoning tasks and does not transfer to perception-bound low-resource ASR.

Anti-repetition decode guards

We tested `no_repeat_ngram_size` and `repetition_penalty` to suppress loops at decode time. Overall WER got worse, from 58.99 to 61.61. The reason is linguistic: these languages use reduplication grammatically, so penalizing repeated tokens suppresses correct output. The guards helped the single worst looper (`nhq`, 100 to 82) but hurt Spanish (18.8 to 20.7) and reduplicating languages (`vmj`, 66 to 78). The conclusion is that "no repetition" must be learned through reinforcement learning, which it was, and not bolted on at decode time.

Continuation RL

Running more GSPO from the already-tuned model, with a lower learning rate and a stronger anti-repetition reward, never beat its own starting point: dev CER stayed between 0.43 and 0.44 against a 0.414 start. The first GSPO pass extracts the available gain; more of the same kind plateaus.

Balanced data and full-epoch training

We rebuilt the broad manifest with Common Voice capped per language so the 23 test languages carried half the weight rather than a quarter, then trained two full epochs. Both choices were wrong. Full training overfit and pushed the loop rate to 23.9% and WER to 123, and reducing the transfer data made the base weaker. This confirmed, a second time, that the transfer data helps and that early stopping is essential.

Engineering and infrastructure

Two engineering lessons cost real time and are worth recording.

The first is a five-hour GPU idle caused by a process-detection bug. A chain of background watchers used `pgrep -f PATTERN` to decide whether a training run was still alive. `pgrep -f` matches the full command line of every process, including the watcher's own inline command and any concurrent status-check command that contained the same pattern, so the watchers kept seeing a finished run as alive and never launched the next stage. The fix is to detect completion through an explicit `EXIT` marker written to the log file (`grep -q "^EXIT"`), never through process-name matching, and to trust `nvidia-smi --query-compute-apps` for "is anything on the GPU."

The second is operational: the cloud machine is an ephemeral container whose `/tmp` is wiped on restart, which once silently destroyed a staged launcher. All launchers, state, and persisted models now live under a durable path, and the conversion and evaluation tooling was hardened against three Transformers 5.5 quirks (a stub tokenizer, a `processor_config.json` versus `preprocessor_config.json` naming change, and a `dtype` versus `torch_dtype` config key).

Key lessons

1. Teacher-forced validation metrics lie for this task. Select checkpoints by autoregressive evaluation with an explicit loop metric.
2. Data scale reduces repetition (a measured 26.3% to 15.9% loop rate at three times the data), but reinforcement learning is what actually closes the WER gap.
3. Supervised fine-tuning must stop early. Every full-epoch run overfit and amplified looping.
4. "No repetition" is a training property, not a decode-time patch, because these languages use grammatical reduplication.
5. Recipes designed for reasoning models (frontier weighting, self-distillation, chain of thought) assume structure that low-resource perceptual ASR does not have.

Limitations and future work

The honest ceiling is data. At about ten hours per language, the six to eight hardest languages dominate the average and cannot be modeled into shape. Average WER of 20 is not reachable at this data scale. The model also produces no timestamps, operates on segments under 30 seconds, and decodes everything through a single `es` bucket rather than a true language identifier.

The documented next levers, both larger investments, are a higher-capacity backbone (whisper-large-v3 with its 32-layer decoder, traded against slower reinforcement learning), and targeted per-language data collection for the failing languages, including the large untranscribed Nahuatl deposits that could be pseudo-labeled. A faster reinforcement-learning loop using accelerated sampling would help exploration but would not raise the data ceiling.

Conclusion

NeblinIA-Speech preview-0.1 is a real, top-ranked foundational ASR model for 23 Mexican Indigenous languages, improving the fair-evaluation WER from 66 to 59 through a combination of open multilingual data scaling and verifiable-reward reinforcement learning. The result is bankable and usable, and the surrounding catalogue of what failed, and precisely why, is intended to save the next effort from repeating the same dead ends. The languages that remain hard are limited by data, not by the method, which sets a clear agenda for the next round.